On Compensator Design for Linear Time-Invariant Dual-Input Single-Output Systems

Steven J. Schroeck, William C. Messner, *Member, IEEE,*, and Robert J. McNab

***Abstract*—This paper presents a new method for the design of compensators for linear time-invariant dual-input/single-output (DISO) systems in continuous time or discrete time. The new method reduces the problem to two single-input/single-output (SISO) design problems, which are well suited to frequency-response design techniques. The first part of the method is the design of a stabilizing compensator for an auxiliary feedback system. The auxiliary compensator parameterizes the two output blocks of the single-input/dual-output compensator such that the zeros of the parallel system formed by cascade of the compensator with the plant are stable. The auxiliary compensator also determines the relative contribution to the output of the two parallel subsystems of the DISO system. The second SISO compensator design is used to ensure that the feedback system is stable and that performance and robustness specifications are achieved. This paper includes a discrete-time design example for a dual-stage actuator system for a disk drive including implementation results. Straightforward extensions for multi-input/single-output systems are discussed.**

***Index Terms*—Linear systems, multiple-input/multiple-output (MIMO) control, parallel systems, two-stage actuators.**

I. Introduction

MANY position servo systems contain two (or more) actuators that act in parallel. One low-bandwidth large-stroke (coarse position) actuator serves to move a high-bandwidth short-stroke (fine position) actuator, which in turn moves whatever is to be accurately positioned. An example of a dual-input single-output (DISO) system is the coarse and fine actuators of an optical disk drive.

Recently, there has been increased interest in the development of more sophisticated control for DISO systems with the increasing bandwidth requirements for magnetic hard disk drives and optical disk drives. For hard disk drives, a variety of industry and university research groups have produced second-stage actuator prototypes to supplement the traditional voice coil motor actuator [1]–[6].

DISO systems are a subset of multi-input/multi-output (MIMO) systems, and thus the servo engineer can apply the design methods developed for MIMO compensator design to the DISO compensator problem. These techniques include $H2, H_\infty$, and μ-synthesis [7]. Disadvantages of these methods are that they are usually best applied to continuous-time systems, and rather high order compensators are often the result of the design process.

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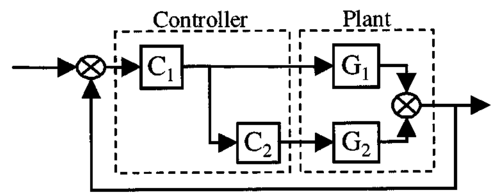


Fig. 1. Typical master–slave structure.

The master–slave method is an attempt to treat the MIMO compensator design as a sequence of single-input single-output (SISO) designs under the assumption that there is very little interaction between the SISO loops. The structure uses the output of the fine actuator compensator as the reference input to the coarse actuator compensator, as shown in Fig. 1. The idea is that the output of the fine actuator compensator C_1 is a good estimate of the relative position of coarse and fine actuator at low frequencies, and the coarse actuator control loop attempts to keep that relative position at zero. Unfortunately, there is usually more interaction between the DISO subsystems than the approximation warrants, and the performance of the closed-loop system suffers.

The approach introduced in this paper takes advantage of the fact that the structure of a DISO system is significantly simpler than the structure of the general MIMO system. While the method involves the design of two SISO compensators, it is very different from the master–slave technique.

This paper is organized as follows. Section II discusses issues in compensator design for DISO systems. Section III develops the new design method, and Section IV shows an application to the design of a compensator for a dual-stage actuator system for hard disk drives in discrete time. Section V shows how to extend the method to the design of compensators for general multi-input/single-output systems, and Section VI contains concluding remarks.

II. Issues in Compensator Design for DISO Systems

All linear time-invariant feedback systems consisting of a DISO system with a single-input dual-output (SIDO) compensator can have the block diagram form shown in Fig. 2. This

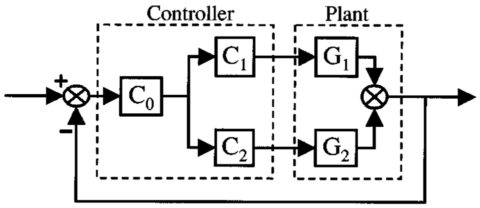


Fig. 2. DISO system block diagram.

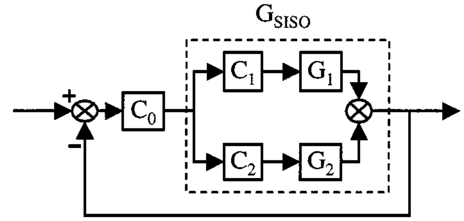


Fig. 3. Depiction of G_{SISO}.

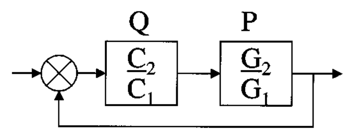


Fig. 4. PQ feedback system.

representation is convenient for the discussion of issues in compensator design for DISO and for the design technique developed in the following section.

The primary issues in the design of any compensator for a control system are the stability and performance of the closed-loop system. Complicating the design of SIDO compensators for DISO systems are the zeros created by the parallel combination of G_1C_1 and G_2C_2. The zeros are the roots of the equation

$$G_1C_1 + G_2C_2 = 0. \tag{1}$$

Stability of the zeros is important because unstable zeros limit the bandwidth of the closed-loop system.

Closely related to the stability of the zeros are the relative contributions to the output from the G_1 and G_2 subsystems as a function of frequency. Assuming, without loss of generality, that G_1 and G_2 represent the dynamics of a fine actuator and coarse actuator, respectively, the output magnitude of G_1 should be large with respect to G_2 at high frequency. Similarly, the magnitude of the output of G_2 should be large with respect to G_1 at low frequency.

The relative phase of the output of G_1 and G_2 is critical at the frequencies where their output magnitudes are nearly the same. If the relative phase is approximately 180° at those frequencies, then the outputs are almost completely out of phase. The subsystems essentially fight each other, and the total output magnitude will drop considerably due to output interference. In such a situation, there is a pair of complex conjugate zeros near the $j\omega$ axis.

III. PQ Design Method

The new design method, which we call the PQ method for reasons that will become clear shortly, consists of two parts. In the development, the following assumption is critical: G_1 represents the dynamics of the high-bandwidth actuator, and G_2 represents the dynamics of the low-bandwidth actuator. The first part is the simultaneous selection of C_1 and C_2 to address the issues of stable zeros, relative output contribution, and destructive interference. The second part is the design of C_0. Once C_1 and C_2 have been selected, the design of C_0 is equivalent to the design of a compensator for the SISO system in Fig. 3 where

$$G_{\mathrm{SISO}} = C_1G_1 + C_2G_2. \tag{2}$$

The selection of C_1 and C_2 is critical, because these transfer functions determine the zeros of G_{SISO} through (1) and the ratio of output contributions from G_1 and G_2 through the relation

$$\frac{G_2C_2}{G_1C_1} = PQ, \quad \text{where } P = \frac{G_2}{G_1} \quad \text{and} \quad Q = \frac{C_2}{C_1}. \tag{3}$$

Since G_2 represents the dynamics of the coarse (low-bandwidth) actuator and G_1 represents the dynamics of the fine (high-bandwidth) actuator, then the product PQ should have large magnitude $(\gg 1)$ at low frequency and small magnitude $(\ll 1)$ at high frequency. Large-magnitude PQ implies a large relative contribution from the coarse actuator, and small-magnitude PQ implies a large relative contribution from the fine actuator. When the magnitude of PQ is nearly one (0 dB), the outputs from G_1 and G_2 have nearly the same magnitude. The phase of PQ determines the relative phase of the outputs of G_1 and G_2, and thus how much the two subsystems destructively interfere.

The problem of selecting C_1 and C_2 becomes a SISO design problem in the following way. Dividing (1) by G_1C_1 leads to the relation

$$1 + \frac{G_2C_2}{G_1C_1} = 1 + PQ = 0. \tag{4}$$

Therefore, the roots of (4) are the zeros of the parallel subsystem G_{SISO} of (2). The key observation at this point is that (4) is the characteristic equation of the PQ feedback system of Fig. 4. Thus, selecting Q to stabilize the PQ feedback system ensures that G_{SISO} will have stable zeros, except possibly for those that are common to G_1 and G_2.

Frequency-response techniques are most suitable for the design of Q, because Q directly influences the product PQ, which determines the relative contribution of the G_1 and G_2 subsystems as a function of frequency. The 0-dB crossover of PQ is the frequency at which magnitudes of the output contributions of G_1 and G_2 are the same. Through the choice of Q, this frequency is at the discretion of the servo designer.

Assuming that P is realizable and has stable poles, the phase margin of the PQ feedback system is the difference between $-180°$ and the phase of PQ at the 0-dB crossover frequency.

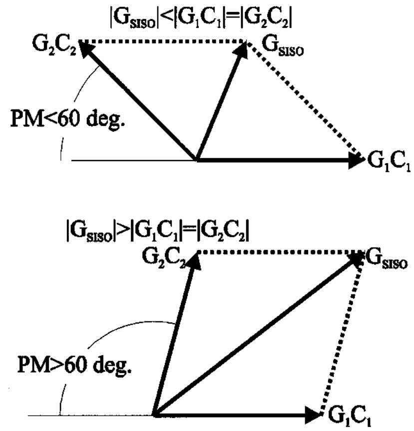


Fig. 5. Effect of phase margin on $G_{\rm SISO}$ magnitude.

This fact implies that the phase margin of the PQ feedback system determines both the relative stability of the zeros of $G_{\rm SISO}$ and the amount of destructive interference between the G_1 and G_2 subsystems. Fig. 5 shows that, for phase margins greater than 60°, the magnitude of the output of $G_{\rm SISO}$ will be greater than one.

After selecting Q, the servo designer must select both C_1 and C_2 to be realizable and to satisfy (3). To keep the order of the SIDO compensator low, the designer should select C_1 to be as simple as possible. If Q itself is realizable, then $C_1 = 1$ is a reasonable choice. The servo designer can remove any factors common to C_1 and C_2 and insert them into C_0 or eliminate them.

At this point, the method for the design of the SIDO compensator for the DISO plant proceeds as a standard SISO design problem for choosing C_0 for the SISO plant $G_{\rm SISO}$. Many SISO design methods are at the disposal of the servo engineer, but once again frequency-response methods are likely to be most useful for achieving bandwidth, disturbance rejection, and robustness specifications.

If $C_1 = 1$, the structure of the PQ design is the same as the master–slave structure in Fig. 1. However, the PQ design is likely (but not guaranteed) to have better performance than a design devised using the typical master–slave technique. The PQ method explicitly addresses relative output contribution as a function of frequency and the stability of the zeros of the parallel loop, but the master–slave technique does not address these issues directly. Designers more comfortable with the master–slave technique might still find the PQ method to be valuable as an analysis tool.

IV. Discrete-Time Compensator Design Example

This section demonstrates the design and implementation of a discrete-time compensator for a dual-stage actuator test apparatus at Western Digital Corporation. Fig. 6 shows a schematic. The first stage consists of a conventional voice coil motor (VCM) with pivot bearing and E-block. The second stage consists of a modified suspension developed by Hutchinson Technology Incorporated, to which lead-zirconium-titanate (PZT) piezoelectric strips have been attached [2]. Energizing the PZT strips produces small displacements of the head relative to the base-plate of the suspension, which is attached to the E-block.

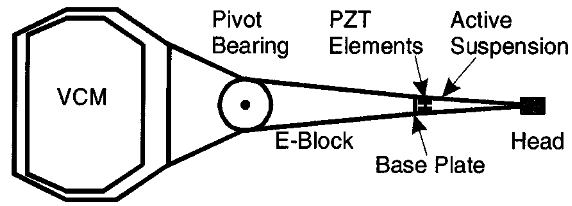


Fig. 6. Dual-stage HDD actuator.

One advantage of the PQ design method is that direct design of discrete-time compensators is straightforward. Fig. 7 shows the experimentally obtained frequency responses for G_1, the transfer function from the PZT input to head motion output, and, for G_2, the transfer function from the VCM input to head motion output, respectively. The sampling rate is 20 kHz. The design objective is to achieve at least a 1500 Hz, 0 dB crossover with at least 35° of phase margin for $G_{\rm SISO}C_0$.

The Bode plot of the ratio $P = G_2/G_1$ is shown in Fig. 8(a). We chose a 250 Hz, 0 dB crossover based on estimates of the runout of the rotating disk at that frequency, and saturation limits of the PZT actuator. The crossover frequency should be low to achieve large error rejection contribution from the fine stage, but not so low that the fine stage is exceeds it range. To assure that all zeros are stable, we select a 65° phase margin for the PQ system. The rate at which the handoff occurs between the two stages depends on the slope of the PQ magnitude plot. Further phase margin increases would improve cooperation between the stages but would flatten the slope of the magnitude curve as it passes through 0 dB. To prevent saturation of the fine stage, the crossover frequency would then have to be increased.

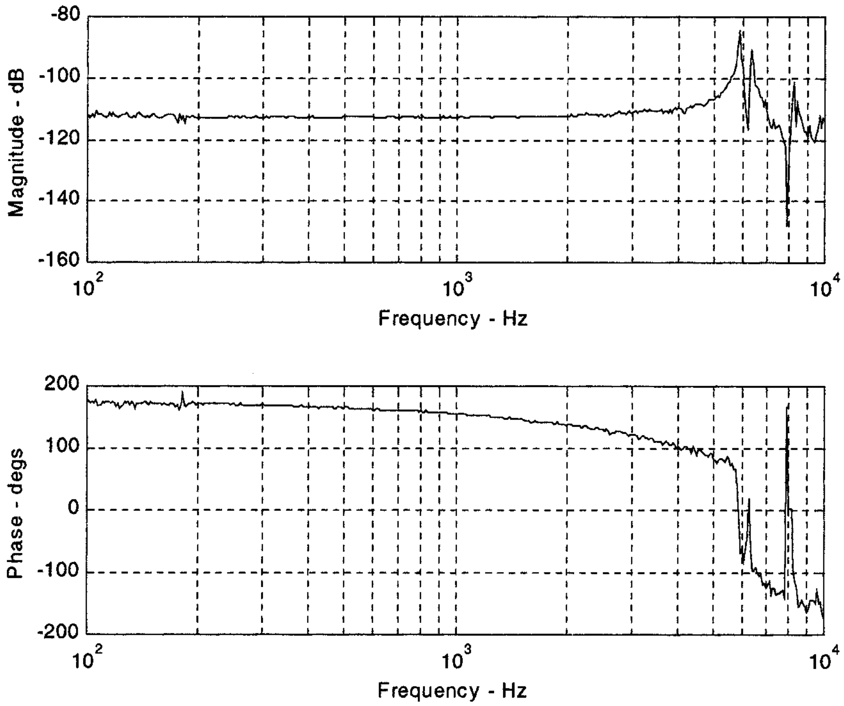
The transfer function of Q is

$$Q(z) = 5\left(\frac{z - 0.986\,936}{z - 0.619\,711}\right)\left(0.001\frac{3.358z - 3.308}{z - 1}\right)$$

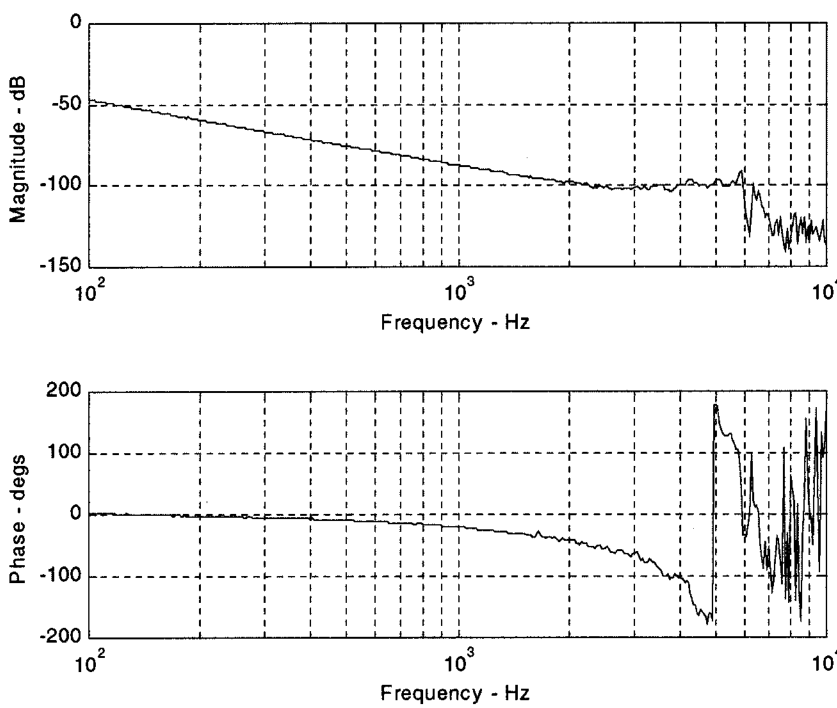
which is a lead compensator to increase the phase at the crossover frequency along with an integrator for low-frequency disturbance rejection by the VCM. A zero is included with the integrator to reduce the phase loss as the frequency increases. Fig. 8(b) shows the Bode plot of PQ. Since Q is realizable, $C_1 = 1$ and $C_2 = Q$.

Fig. 9(a) shows the Bode plot of $G_{\rm SISO} = G_1C_1 + G_2C_2$. The objective for the compensator for $G_{\rm SISO}$ is to reduce the resonance peaks and boost gain at low frequencies. Based on these criteria, we choose $C_0(z)$ to be

$$\begin{aligned} C_0(z) = -3.45e5 &\left(\frac{z - 0.087\,797}{z - 0.971\,347}\right) \\ &\times \left(\frac{0.467\,527z^2 + 0.294\,529z + 0.467\,527}{z^2 + 0.294\,529z - 0.064\,947}\right) \\ &\times \left(\frac{0.903\,207z^2 + 1.525\,207z + 0.903\,207}{z^2 + 1.525\,207z + 0.806\,415}\right). \end{aligned} \tag{5}$$



(a)



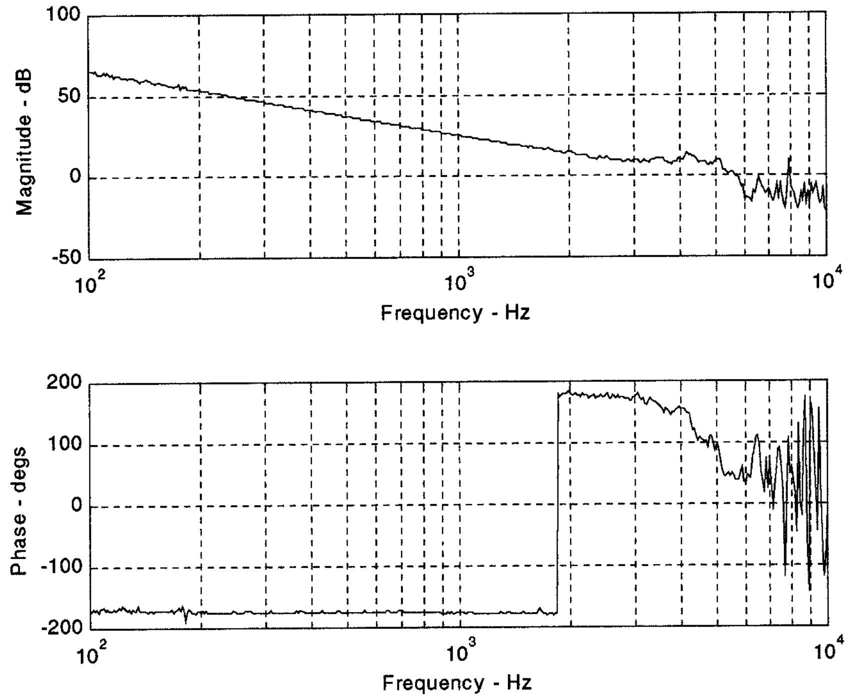
(b)

Fig. 7. Frequency response of (a) G_1 and (b) G_2.

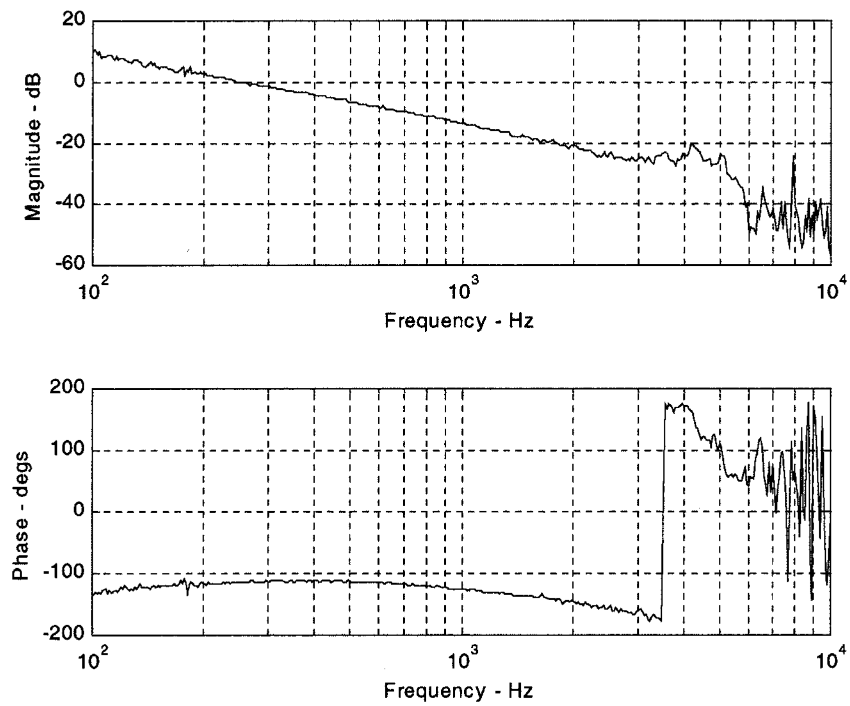
This compensator is the cascade of two notch filters and a lag compensator. The lag filter gives up phase for higher gain at low frequencies. Fig. 9(b) shows the Bode plot of $G_{\mathrm{SISO}}C_0$. The 0-dB crossover is 2 kHz with a phase margin of 35°, and the gain margin is 6 dB.

Fig. 10(a) shows the complementary sensitivity function. Fig. 10(b) shows the sensitivity functions for both dual- and single-stage control. The 0-dB crossover for the dual stage sensitivity function is 1470 Hz and the maximum peaking is 8.9 dB versus 380 Hz and 6 dB for single-stage control. The position error signal (PES) variance from a single-stage drive would be reduced by 48% with the implementation of this dual-stage compensation.

This design balances the tradeoffs inherent in the PQ method to meet the design objectives. The error rejection curve provides



(a)



(b)

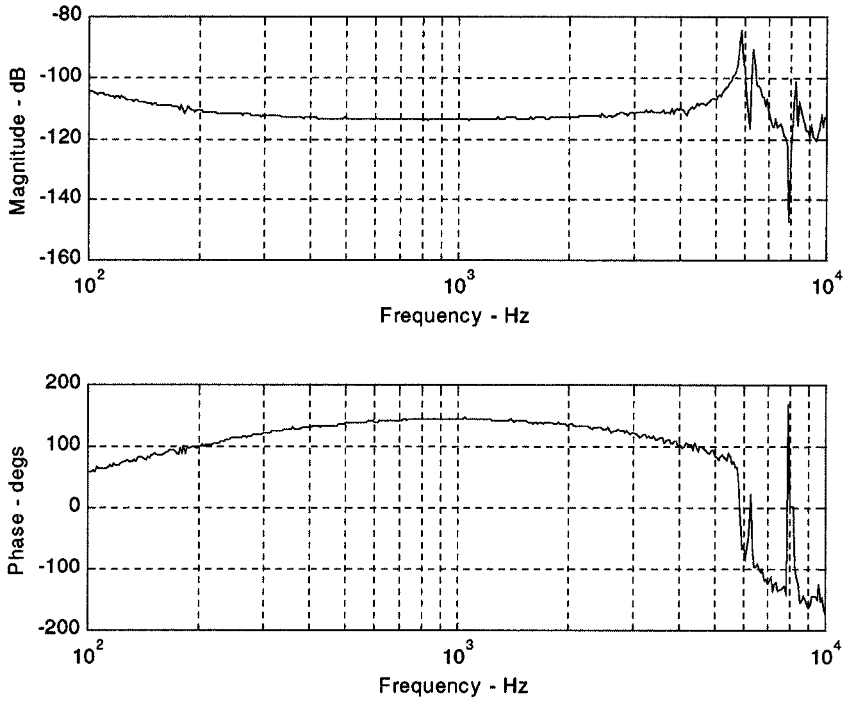
Fig. 8. Bode plot of (a) P and (b) PQ.

good performance in the mid-frequency range (400–1000 Hz), although the peaking (gain above 0 dB) between 2 and 4 kHz may be a concern. Very importantly for implementation in a disk drive, the overall compensator order is low.

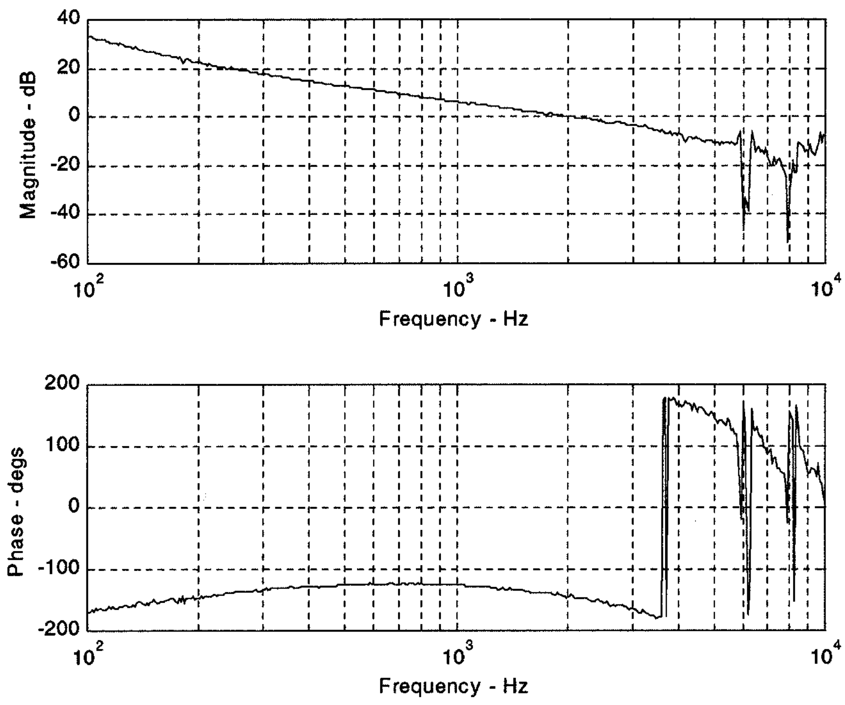
The sensitivity (position error rejection) can be tuned for optimal performance based on the disturbance sources. If the disturbances have a large amount of energy in the same frequency range where there is peaking, then the design should be altered to shift some of the error rejection from the low frequencies to the frequencies of the disturbances.

V. Extension to MISO Systems

Extending the PQ method to the general problem of compensator design for linear MISO systems is easy



(a)



(b)

Fig. 9. Bode plot of (a) $G_{\rm SISO}$ and (b) $G_{\rm SISO}C_0$.

by iteratively applying the method developed for DISO systems. Fig. 11 illustrates the process for an n-input one-output system, which is described in pseudocode below. To facilitate the design process, the ordering of the subsystems G_i should be from higher to lower bandwidth for increasing i, i.e., G_1 has the highest bandwidth and G_n has the lowest.

The extended PQ method designs $(2n-1)$ compensators, which could result in a large redundancy in the controller. However, many of these compensators are likely to be one or constant gains, and therefore the order of the overall compensator can in fact be quite low. Postdesign model reduction could further reduce the redundancy. The utility of the approach will no doubt depend on the system to which it is applied.

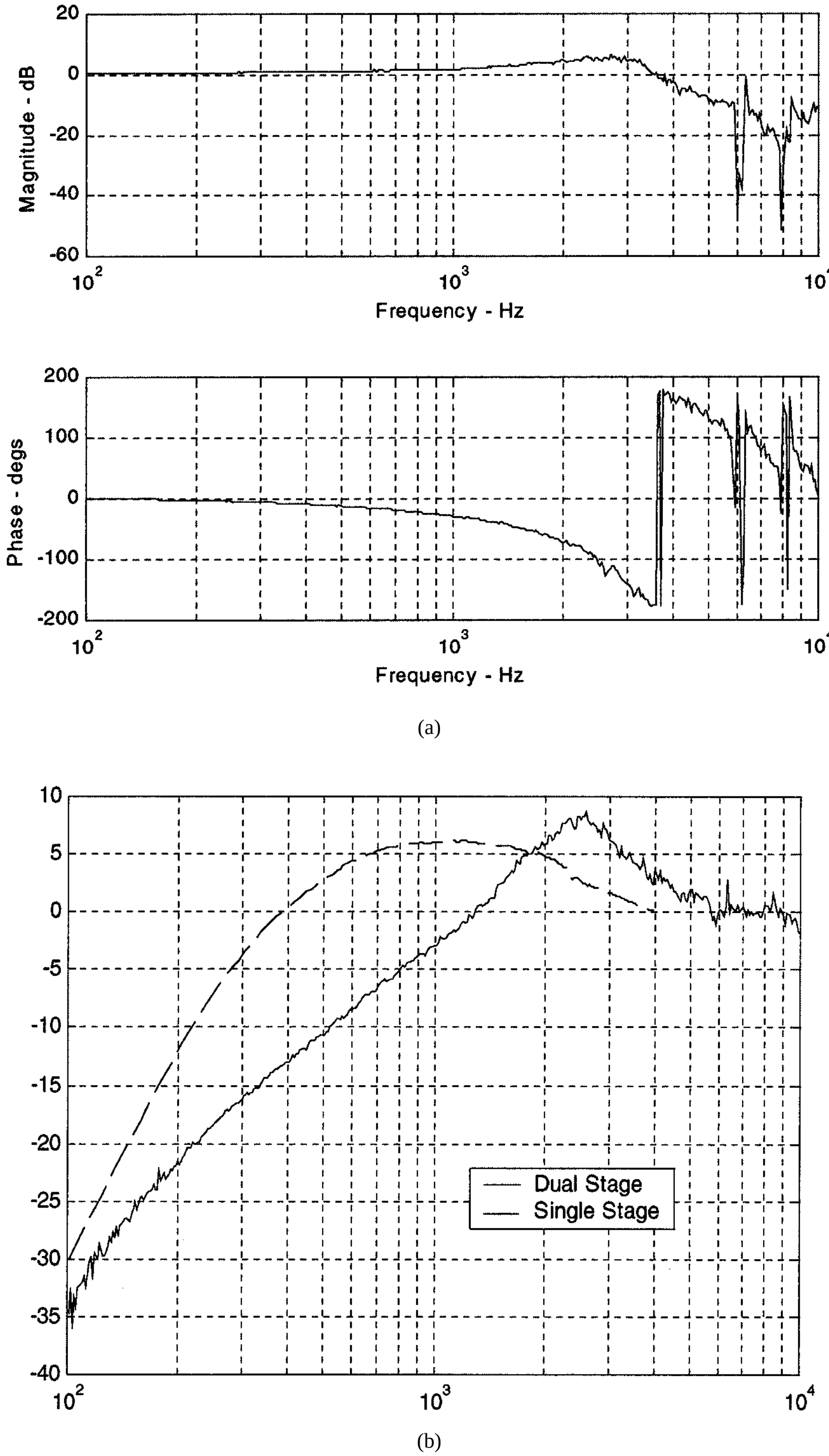


Fig. 10. (a) Complimentary sensitivity function. (b) Sensitivity function.

VI. Conclusions and Future Work

The PQ method is a straightforward, intuitive framework for the design of compensators for linear time-invariant MISO systems and is especially well suited to DISO designs. Within its scope of application, the PQ method produces results as good as or better than more general MIMO design methods with low-order compensators. In addition, direct design of discrete-time compensators is easy with this method. The design example for a hard disk drive system incorporating a dual-stage actuator demonstrated the effectiveness of the PQ method.

There are several avenues for future work on the PQ method. One area is understanding more clearly the impact of the PQ feedback gain and phase margin on the total system robustness. A second area is the development of advanced feedforward methods that exploit the two parallel systems for improved step response and rejection of periodic disturbances while the entire system is under the feedback control designed by the PQ

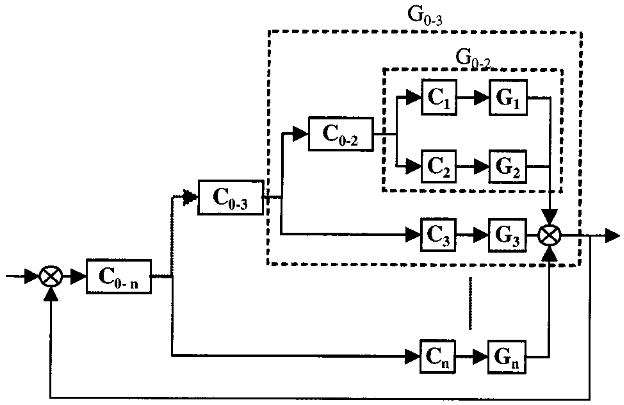


Fig. 11. General MISO system.

method. Yet another area is extending the PQ method for application to certain types of SIMO in which one output is to be controlled and the others serve as feedback to improve performance.

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